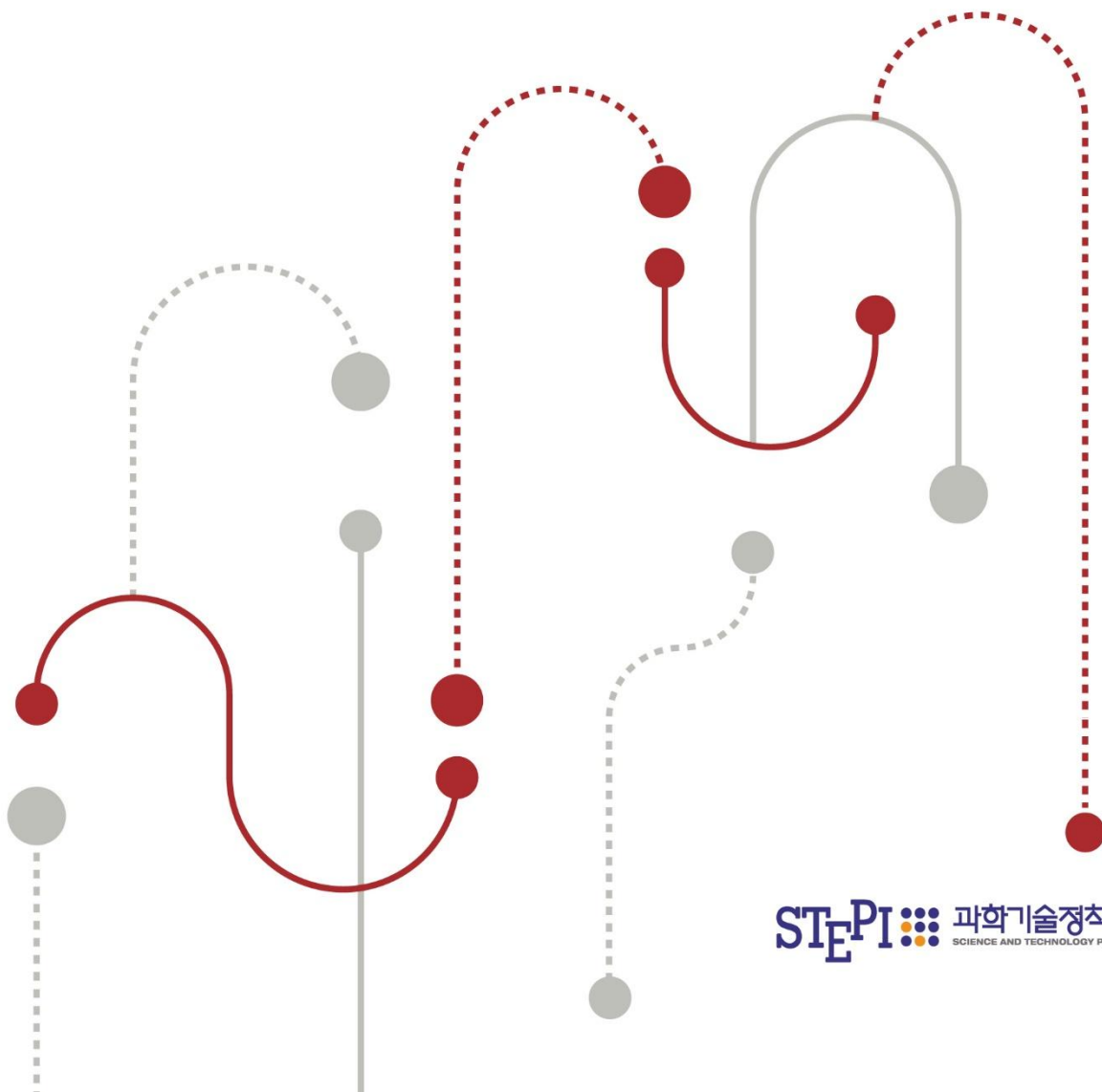


SCIENCE & TECHNOLOGY POLICY

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A Framework for Legislative Impact Assessment For the Science and Technology Innovation Transition



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**A Framework for Legislative
Impact Assessment For the
Science and Technology
Innovation Transition**

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SUMMARY

A Framework for Legislative Impact Assessment For the Science and Technology Innovation Transition

Jieun Jeon

■ Research Background: Changes in the Science and Technology Innovation Environment and the Need for Legislative Impact Assessment

- (Transformative nature of S&T innovation) Structural changes across society, economy, and environment → deepening policy and legislative uncertainty
- (Technology convergence and strategic intelligence) Spread of disruptive technologies(AI, Quantum computing, Synthetic biology) and technological convergence, Limitations of prediction-centered, static policy approaches, Need for agile, experimental policy making and strategic intelligence
- (Colligridge's dilemma) (a) Early stage: Easy to control, but effect unknown, (b) Later stage: Effects clear, but technology already entrenched and uncontrollable, (c) Need for proactive discussion and phased institutional design to manage uncertainty
- (Structural conflict) "Definitive norms" pursuing legal stability vs. "fluidity of innovation" → obstruction of innovation or regulatory gaps

■ Research Objectives

- Explore the necessity of legislative impact assessment (LIA) and a framework to address the transformative phase of S&T innovation
- Compare domestic and international LIA practice cases
- Propose Korea-specific S&T LIA system building plan

■ Concept and Significance of Legislative Impact Assessment

- Definition: An evidence-based legislative approach systematically reviewing law effects, evaluating effectiveness and equity, and feeding results back into legislation
- Significance
 - Achieves appropriateness, efficiency, and fairness in legislation
 - Provides evidence base for legislative necessity and policy choices
 - Enables dynamic norm formation: ex-ante ↔ ex-post evaluation cycle → continuous learning and improvement

■ Domestic and International LIA Status

- Korea(Executive): sector-specific ex-ante assessments (regulation, technology, gender, employment, environment)
 - Fragmented by ministry and law → limited integration and post-utilization
- Korea(National Assembly): expert review reports, cost estimation, pilot LIA projects
- International Cases
 - EU: integrated economic·social·environmental assessment / “Better Regulation” package → standardized items
 - US: cost-benefit RIA for significant regulations / independent oversight
 - Switzerland: constitutional basis → effectiveness·efficiency·centered evaluation
 - Germany: GGO-based 3-phase cycle (ex-ante → concurrent → ex-post)
 - Japan: ministry self-evaluation → coordination → public disclosure for feedback
- Key Implications
 - Common elements across countries: problem definition → alternative comparison → effect/side-effect analysis → cost-benefit review

- LIA must extend beyond ex-ante → feedback loop linking post-review, revision, and supplementation
- Three effectiveness factors: standardized criteria / independent expert review / stakeholder consultation

■ Framework for S&T-Specific LIA System

- Basic Direction
 - S&T legislation → shift to dynamic normative framework premised on learning and adjustment
 - Full-cycle LIA: ex-ante → implementation → ex-post → learning & feedback
 - Assessment scope: beyond cost estimation → innovation direction / social acceptability / risk-benefit distribution / long-term transition effects

■ Recommendations

Recommendation 1. Data-driven analytical infrastructure with future forecasting and scenario analysis

- Integrate enforcement outcomes, post-LIA data, government assessment results
- Introduce forecasting, scenario analysis, early warning functions → proactive response to high-uncertainty legislation
- Leverage AI → impact prediction / pattern analysis / stakeholder response detection

Recommendation 2. Institutionalize LIA with ex-ante ↔ ex-post linkage and feedback mechanism

- Legal/procedural foundation: assessment targets, scope, procedures, result utilization, post-review obligations
- Performance indicators → effect verification → revision/repeal decisions
- Common items standardized + flexible S&T-specific additional items

Recommendation 3. Independent expert evaluation + bidirectional legislature-executive feedback governance

- Clear accountability: assessment planning / data collection / analysis / review / post-monitoring
- Independent review body: S&T, economics, society, law, ethics, environment experts
- Legislature ↔ executive bidirectional feedback → mutual utilization of assessment data and results